



NATIONAL UNIVERSITY OF MODERN LANGUAGES ISLAMABAD
FACULTY OF ENGINEERING & COMPUTING
DEPARTMENT OF COMPUTER SCIENCE
END-TERM EXAMINATION – FALL 2025 – BSCS VII A&B (MORNING)

Subject: Software Project Management

Instructor: Dr. Sarah Mazhar

Allowed Time: 03 Hours

Total Marks: 50

Instructions: -

- i. Attempt all questions and return the question paper with the answer sheet.
- ii. Clearly mention the question number before attempting it

QUESTION 1 [CLO-1, C2, PLO-1]

Part 1: Explain the process of project integration management. Moreover, discuss the key outputs of this knowledge area. **[7+2 = 9 Marks]**

Part 2: What is a statement of work (SOW) and in which knowledge area it is developed? **[1 Mark]**

Part 3: Describe the fundamental communication methods and the widely used project communication tools currently prevalent in the industry. **[3+2 = 5 Marks]**

QUESTION 2 [CLO-2, C4, PLO-3]

Analyse the following scenario with respect to project risk management and classify the identified risks into appropriate risk categories. (Any five) **[5*2= 10 Marks]**

A mid-sized software company, SoftLine Solutions, has secured a contract to develop an online examination system for a large educational board. The project has a strict six-month deadline because the system must be deployed before the upcoming exam cycle. The client frequently changes requirements due to evolving government policies. The development team includes several junior developers who have limited experience with secure coding practices. The company is also integrating a third-party biometric verification API, but the vendor has not yet provided stable documentation. Additionally, the organization recently shifted to a hybrid work model (Some employees work from the office, while others work remotely from home), causing occasional communication gaps between team members. The budget is fixed, and any delay may lead to financial penalties.

QUESTION 3 [CLO-3, C4, PLO-4]

Calculate the effort estimate, TDEV, average staffing, and the cost required to develop the product with Constructive Cost Model i.e. with COCOMO 1 and COCOMO 2, for a project with 80 KLOC (thousands of lines of code) and the value of cost drivers (Multipliers) is 1.17. Assume that the average salary of software engineers be \$6000/- per month. **[(3+3+2+2)*2 = 20 Marks]**

QUESTION 4 [CLO-3, C4, PLO-4]

Analyze the project closure scenario provided and determine the essential project closure activities that must be performed to formally complete the project.

A software company, CodeCraft, has delivered an online fee management system to a university. The system has passed User Acceptance Testing, and the university has confirmed that all major features work as expected. However, the project team has not yet handed over the complete user manual, and one final training session for the university staff is still pending. Some project documents, such as the updated requirement list and test summary report, have not been archived. A few vendor invoices are also awaiting approval. The university requires a formal project closure report before releasing the final payment.

[5 Marks]